Domain Types in SQL

Built-in Data Types in SQL

Definition

Domain types define the set of values that an attribute in a database relation can take. SQL supports a variety of built-in domain types suitable for different kinds of data.

Common SQL Domain Types

1. char(n):

- Fixed-length character string.
- Length is specified by the user as `n`.
- Example: `char(10)` always occupies 10 characters, padding with spaces if needed.

2. varchar(n):

- Variable-length character string.
- Maximum length specified by `n`.
- Example: `varchar(50)` can store up to 50 characters without padding.

3. int:

- Integer type.
- Stores a finite subset of integers, machine-dependent.

4. smallint:

- Smaller integer type.
- Stores a machine-dependent smaller range of integers than `int`.

5. numeric(p,d):

- Fixed-point number with precision `p` and scale `d`.
- `p` = total digits; `d` = digits after decimal point.
- Example: `numeric(8,2)` allows numbers like 123456.78.

6. real, double precision:

- Floating-point numbers.
- Machine-dependent precision for real and double-precision values.

7. float(n):

- Floating-point number with user-specified precision `n`.

- Ensures at least `n` significant digits.

Note

Choosing the correct domain type is crucial for **data integrity, storage efficiency, and precision**. For example, `char` is better for fixed codes, while `varchar` saves space for variable-length strings.